

Lecture - 17

Orthonormal vectors $\rightarrow$

$$q_i^T q_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$$\# \quad Q = \begin{bmatrix} 1 & & 1 \\ q_1 & \dots & q_n \\ 1 & & 1 \end{bmatrix}$$

$$\boxed{Q^T Q = I} \quad \text{Always !!}$$

Orthogonal matrix $\rightarrow$ square matrix

orthonormal columns

If Q is square then $Q^T Q = I$
means that
 $Q^T = Q^{-1}$

Eg Perm $Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$$Q^T Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Row ops to make $Q \rightarrow I$

$$Q^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = Q^T$$

$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

$Q = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \sim \sqrt{2} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$

orthogonal matrix

$$\# Q = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

$$\frac{1}{2} \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

orthogonal matrix.

$$\# Q = \begin{bmatrix} 1 & -2 \\ 2 & -1 \\ 2 & 2 \end{bmatrix} \rightsquigarrow \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$$

$\Rightarrow$ Why orthogonal matrices

$$\text{Projection matrix} = Q \underbrace{(Q^T Q)^{-1}}_I Q^T = \boxed{Q Q^T}$$

Always identity!
for orthonormal columns

$$P = Q Q^T$$



I when Q is square

$$P^2 = P$$

$$(Q Q^T)(Q Q^T) \Rightarrow Q \underbrace{(Q^T Q)}_{\underline{I}} Q^T$$

↪ $\boxed{Q Q^T}$

The estimation equation becomes quite trivial when we have orthonormal columns.

$$A^T A \hat{x} = A^T b$$

• If $A = Q$

$$\underbrace{Q^T Q}_{\underline{I}} \hat{x} = Q^T b$$

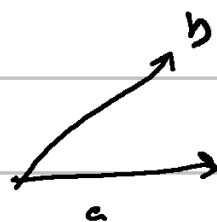
$$\boxed{\hat{x} = Q^T b}$$

$$\boxed{\hat{x}_i = q_i^T b} !!$$

Gram - Schmidt

⇒ When we want to convert a matrix into orthonormal columns matrix

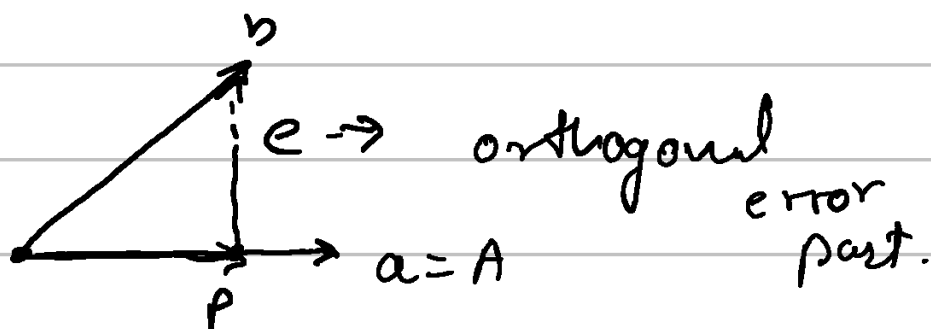
eg independent vector a, b



① The first step is to find two orthogonal vectors A, B from a & b

② orthonormal vectors $\rightarrow q_1 = \frac{A}{\|A\|} \quad q_2 = \frac{B}{\|B\|}$

Let's say $A = a$



$$\begin{aligned}
 B &= b - p \\
 &= b - \frac{A^T b}{A^T A} A \\
 &= b - \frac{a^T b}{a^T a} a
 \end{aligned}$$

Now, $B \perp A$, so $A^T B$ should be zero.

$$a^T \left(b - \frac{a^T b}{a^T a} a \right) = a^T b - \overset{\substack{\text{scalar} \\ \uparrow}}{\frac{a^T (a^T b) a}{a^T a}}$$

$$\Rightarrow a^T b \left[1 - \overset{\substack{\text{scalar too for line}}}{\frac{a^T a}{a^T a}} \right] = \underline{\underline{0}}$$

Now let's say we had another vector $\vec{c}$ and we have to find C which would be our 3rd orthogonal vector.

$$C \perp B$$

$$C \perp A$$

$$\# \quad C = \vec{C} - \underbrace{\frac{A^T C A}{A^T A} - \frac{B^T C B}{B^T B}}_{\downarrow}$$

remove the components
in A & B direction

eg $a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$

Let's say $A = a$

$$B = b - \frac{a^T b}{a^T a} a = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$a^T b = 3$$

$$a^T a = 3$$

$$B = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$q_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}$$

$$\boxed{C(A) = C(Q)}$$

$$\boxed{A = QR}$$

→ Gram-Schmidt
expression

Turns out that R is upper
triangular. ! imp

The idea is how we do the
process $a \rightarrow b \rightarrow c \rightarrow d \dots$

$$\begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ a_1 & a_2 & a_3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} q_1^T a & q_1^T b \\ 0 & q_2^T b \\ 0 & 0 \end{bmatrix}$$

Since we find q_1 only from a

we find q_2 from a & b

$\Rightarrow$ And this algorithm so to say comes out in the form that the relation matrix $\textcircled{R}$ is upper triangular.